

MUSCLE TISSUE

1- INTRODUCTION

MUSCLE TISSUE

Muscle tissue is characterized by the property of contractility, ensured by muscle cells or fibers or myocytes.

These cells are characterized by the presence in their cytoplasm of numerous contractile protein myofilaments.

There are three types of muscle tissue:

- **Striated muscle tissue**, generally associated with the skeleton, is composed of myocytes (rhabdomyocytes) which are responsible for voluntary and rapid movements controlled by the cerebrospinal nervous system.
- **Smooth muscle tissue**, formed of leiomyocytes located in the wall of the viscera and vessels, with involuntary and slow contraction, under the control of the autonomic nervous system.
- **The myocardial tissue**, although striated, is characterized by its ability to contract rhythmically and harmoniously in a spontaneous and involuntary manner.

2- STRIATED MUSCLE TISSUE :

Striated muscles are formed of muscle cells juxtaposed in parallel, in bundles surrounded by vascular connective tissue: the epimysium, another connective tissue the perimysium divides the muscle into bundles, supports the vascular network and surrounds all the nerve elements.

Each muscle fiber is surrounded by connective tissue: the endomysium originating from the perimysium.

2-1- Appearance under optical microscopy:

- Muscle "fibers" are elongated, multinucleated with regular transverse striations
- These cells measure 10 to 100 μm in diameter.
- The length varies from a few hundred μm (eye muscles) to several centimeters (some skeletal muscles).
- Transverse striation results from the organization of myofilaments: myosin and actin.
- The multiple nuclei, arranged against the Plasma membrane, elongated ovoids in the direction of the fiber.
- The plasma membrane is lined by a basal lamina: sarcolemma.
- The sarcoplasm contains myofibrils, mitochondria, and smooth sarcoplasmic reticulum organized in a specific way.
- The myofibrils group together into bundles which, in cross-section, form polygons; between the myofibrils are sarcoplasmic bands.

2-2- In electron microscopy

a) Myofibrils are made up of the juxtaposition of sarcomeres, formed by the highly organized spatial arrangement of myosin and actin.

- Thick myosin filaments:

- Have a diameter of 15 nm and a length of 1.5 μm .
- parallel to each other along the entire length of strip A.

- In cross-section, their regular arrangement forms a hexagon
- In the middle of the A band, the myosin myofilaments are linked by transverse protein bridges corresponding to the M line.
- A myosin filament is an assembly of 300 myosin molecules.
- The myosin molecule is a fibrous, pipe-shaped protein 200nm long and 2nm wide with a head and a stem.

The head and stem junction is mobile and the 300 molecules are assembled in such a way that their stems overlap each other and the heads are oriented outwards, adopting a helical arrangement around the filament axis.

- Thin actin filaments, 1µm long and 5-7nm wide

- Extend on either side of the Z line along the entire length of band I.
- They penetrate into band A, more or less deeply depending on the degree of contraction.
- Their ends mark the limits of the H band - The actin filament has a complex molecular structure comprising; actin, tropomyosin and troponin.

Actin is present in filamentous form (F-actin), composed of two polypeptide chains in a double helix.

Within each groove of the F-actin double helix are lodged tropomyosin filaments 40 nm long extending over 7 G-actin molecules.

Each tropomyosin molecule is associated with a troponin molecule.

Composed of 3 subunits:

- (TnT), the troponin binding,
- (TnC) binds calcium
- (TnI) inhibits the interaction between actin and myosin.
- The Z line corresponds to the junction zone of the thin filaments of two neighboring sarcomeres.

At this level, the myofilaments are connected to each other by bridges of intermediate filaments.

In cross-section, each thin filament of a sarcomere is located at the center of a square formed by the four filaments of the neighboring sarcomere.

- **Non-contractile proteins** maintain the arrangement of myofilaments, allowing them a certain degree of elasticity.

- The main proteins are titin (a long fibrous protein running along the sarcomere between the thick and thin filaments and attached at both ends to the Z-lines)

- Peripheral myofibrils also adhere to the plasma membrane via dystrophin.

- Desmin is also found, which ensures lateral cohesion between myofibrils.

- Nebulin, associated with the actin microfilament, ensures the maintenance of the helical shape.

Plasma membrane and T-tubules

- At each A-band - I-band junction, the plasma membrane forms tubular invaginations that penetrate the cell, bypassing the myofibrils and forming an intracellular tubular network.

- Each T-tubule is flanked on either side by a cisterna of the smooth endoplasmic reticulum, thus forming triads,

- T-tubules serve to bring the depolarization of the plasma membrane as close as possible to the cisternae of the endoplasmic reticulum during nerve excitation, thus promoting the rapid release of calcium and muscle contraction.

Cytoplasmic organelles

Mitochondria aligned between myofibrils, Golgi cisternae, lipid droplets, glycogen and myoglobin.

2-3- Muscle contraction

Muscle contraction results in a shortening of muscle cells, which is reflected microscopically by a shortening of the I bands, while the A bands maintain a constant length .

- It results from the active sliding of thick myosin filaments between thin actin filaments.
- The nerve impulse causes a depolarization of the plasma membrane which extends along the membranes of the T system, then it is transferred to the reticulum via the triads.
- The release of Ca^{++} which activates muscle contraction.
- Actin myofilaments slide between myosin myofilaments controlled by myosin heads which bind to and detach from the actin molecule and thus "walk" on the actin filaments.
- The movement of myosin onto actin is possible thanks to the hydrolysis of ATP molecules.
- Muscle contraction is regulated by molecules associated with the actin molecule; at rest, tropomyosin masks the actin-myosin binding site; the release of this site is influenced by Ca^{++} ions initially contained in the cisternae of the endoplasmic reticulum sarcoplasmic.

2-4- Two types of striated muscle cells:

Red fibers (type I): large, rich in myoglobin, mitochondria and redox enzymes.

- Significant ATPase activity.
- These fibers contract slowly in muscles subjected to slow contractions and prolonged (back muscles).

White fibers (type II): less rich in myoglobin, mitochondria and redox enzymes than the previous ones, resulting in rapid contractions and greater fatigue.

2-5- Innervation

a) Motor innervation is provided by motor neurons in the anterior horn of the spinal cord (γ motor neuron) and the brain.

- Each neuron innervates several muscle fibers.
- Each of them receives a terminal nerve fiber at a particular area, the motor endplate.
- The muscle cell and its motor fibers form a motor unit.
- At the level of the plaque, the sarcolemma forms numerous grooves (synaptic channels) into which the presynaptic nerve endings, rich in synaptic vesicles filled with acetylcholine, sink.
- The axolemma and the sarcolemma remain separated by a synaptic cleft of approximately 50 nm.
- Beneath the sarcolemma are grouped numerous nuclei and mitochondria. Together they constitute the subneural apparatus.
- Nerve excitation triggers the release of acetylcholine into the synaptic cleft and then its binding to specific receptors on the postsynaptic membrane.

- This results in a change in ionic permeability and the appearance of an action potential which propagates along the sarcolemma and leads to the T-tubules.

b) Sensory innervation is provided by nerve fibers either free in the vicinity of muscle fibers, or around modified muscle cells, specific sensory devices:

- Neuromuscular spindles provide information about the body's position in space and contribute to the development of balance reflexes and the regulation of muscle tension.
- Small fibrous bodies, made up of so-called intrafusal muscle fibers, surrounded by a connective tissue capsule.
- Each muscle fiber receives nerve endings from type Ia or IIa fibers.
- These detect the length of the muscles and their tension, and transmit the information.
- Other nerve fibers terminate in the tendon insertion areas and form the neuro-tendinous organs, responsible for tendon sensitivity.

3- CARDIAC MUSCULAR TISSUE

- The myocardium is made up of myocardial muscle cells, differing from the previous ones in certain respects:
- Cardiac muscle cells are mononucleated or binucleated.
- They are much shorter than striated muscle cells.
- They are anastomosed and linked by junction systems, forming a wide mesh network.

Structure of the myocardial cell

A. Optical Microscopy

- The cardiac muscle cell measures 15 to 20 μm in diameter and approximately 100 μm in length. It has one or two central nuclei and is surrounded by a sarcolemma.
 - They are elongated and join together to form anastomosing trabeculae separated from each other by highly vascularized connective tissue.
 - Modified myocardial cells that constitute the nodal tissue.
 - Myocardial cells exhibit striation identical to that of striated muscle due to the presence of actin and myosin myofilaments, organized into sarcomeres.
- There are also transverse densifications: Eberth's scalariform traits which correspond to the junction systems linking the ends of the cells together.

B-Electron Microscopy

- The majority of the sarcoplasm is occupied by the myoplasm organized into a contractile mass. Formed by incompletely separated myofibrils from numerous small mitochondria.

- **Scalariform features** : intercellular junction lines "stair step" with a succession of longitudinal and transverse segments; the actin filaments terminate at this level and associate with the sarcolemma forming an adherent junction.

- The longitudinal part of the scalariform line, communicating junctions nexus which facilitate the passage of membrane excitation.

- **The T-shaped system** :

- In relation to the Z lines, the sarcolemma invaginates to give rise to T tubules that are much wider than in the striated cell.

They are connected to each other by longitudinal tubes and associate with the tubules of the sarcoplasmic reticulum, which do not have terminal cisternae; thus they form diads

- **Nodal tissue** : modified muscle cells that generate and conduct action potentials.

The cells are thin and contain one to two central nuclei surrounded by a clear cytoplasmic zone rich in mitochondria and glycogen.

- The contractile mass is poorly developed

C-Muscle contraction and innervation

- Cardiac muscle contraction is automatic, involving vegetative , controlled by the nervous system nodal tissue and Ca^{++} ions.
- The propagation of the contraction wave is ensured by the nexus-type junctions of the scalariform features.
- The heart is richly innervated by parasympathetic and orthosympathetic plexus endings, important at the base of the heart and between the atria.

4-SMOOTH MUSCLE TISSUE

- Smooth muscle cells possess myofilaments, which are less organized than those of striated muscles.
- They are grouped into bundles to form the muscular tunics of hollow organs (digestive system, urinary tract, genital systems...) and the walls of blood vessels.
- They are subject to slow and sustained contractions, not controlled by the will.

4-1-Structure of the smooth muscle fiber

a) In Optical Microscopy

- The smooth muscle cell is spindle-shaped with a swollen cell body and two tapered ends.
- Its length varies from 15 μ m (at the level of small blood vessels) to 500 μ m (at the level of the uterus).
- Each cell has a central nucleus with an elliptical shape.
- It is surrounded by the cell's organelles, including numerous mitochondria.
- Each cell is surrounded by the sarcolemma formed by the sarcoplasmic membrane and the basal lamina and contains myofilaments oriented along the long axis of the cell.
- Other molecules include calponin and caldesmon in the smooth muscle cell.
- Calponin, a molecule related to troponin(I), binds to F-actin, modifying its conformation and inhibiting the ATPase activity of myosin.
- Thick myosin myofilaments that differ from those of striated muscle cells.
These filaments are unstable and would only form when the fiber undergoes excitation, through polymerization of dispersed myosin molecules.
- Thick myofilaments are much less numerous than in striated cells (1 for every 12 thin myofilaments)
- The sarcolemma is the site of numerous invaginations equivalent to triads.
- Smooth muscle fiber, on the other hand, has a cytoskeleton made up of intermediate filaments with a predominance of desmin in the tunics of the viscera and vimentin at the level of the blood vessels.
- The sarcoplasmic reticulum is poorly developed.
- The contraction of smooth muscle tissue is under the control of the sympathetic and parasympathetic systems.
- No well-defined neuromuscular junction.
- The neurotransmitter is released from the swellings along the axon

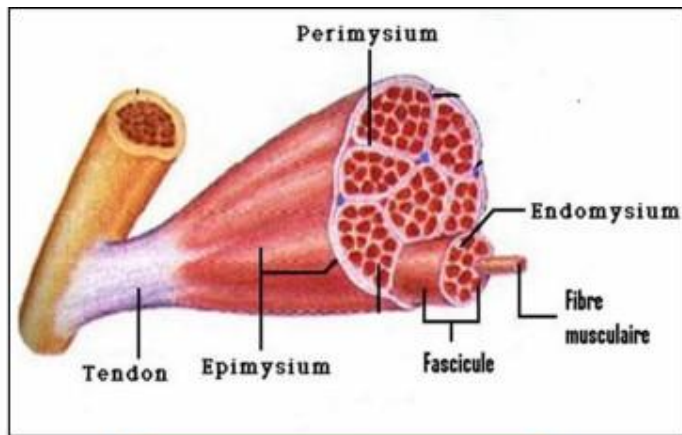
- The contraction is different from that of striated muscle, dependent on Ca^{++} but the control of calcium movements is different from striated muscle.
- Contraction produces a shortening of the cell which takes on a globular shape.
- During the phase of maximum contraction, the nucleus is often folded back on itself.
- The Ca^{++} released at the time of excitation of the cell membrane binds to a calmodulin protein; the complex thus formed activates an enzyme located on the light chain of myosin, which can then bind to actin.
- The two proteins can then interact in a manner identical to what happens in striated muscle.

Depending on the location of the smooth muscle, several varieties of smooth muscle fibers are distinguished: **Visceral mvocytes**: these are the smooth muscle cells that form the walls of hollow organs (digestive tract, urinary excretory pathways, uterus).

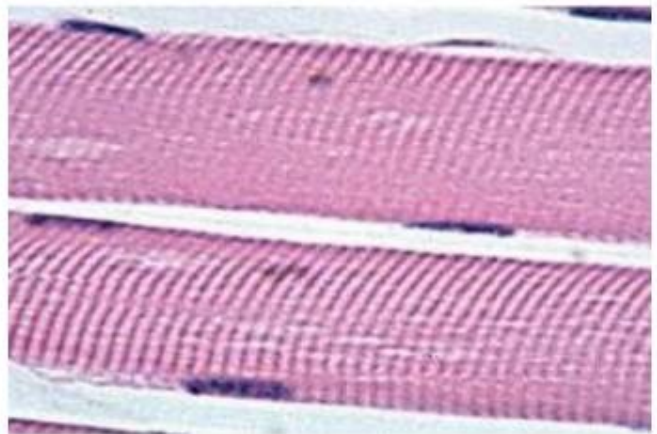
Mvocytes of the vascular walls : different from a morphological point of view: they are cells with branched ends allowing their insertion on the elastic laminae of the arterial walls. **Pericytes**: poorly differentiated, spindle-shaped, around capillaries and venules they are surrounded by a basal lamina and possess filaments of actin and myosin.

Mvoepithelial cells: flattened, containing contractile proteins and desmin intermediate filaments.

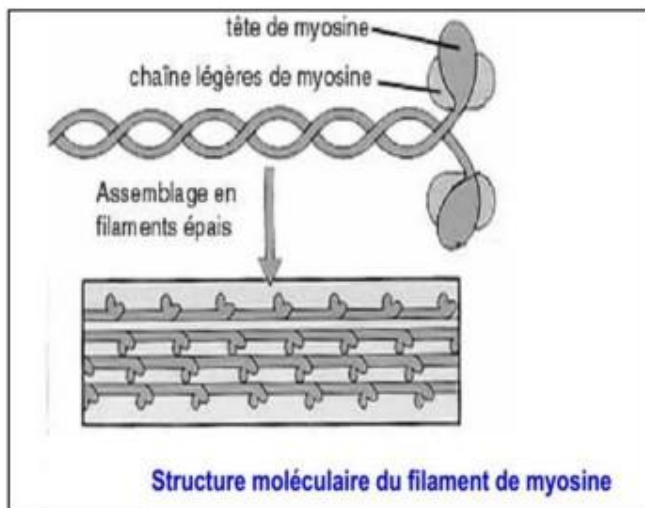
Mvofibroblasts: spindle-shaped, similar to fibroblasts, they secrete collagen but have contractile properties thanks to actin.



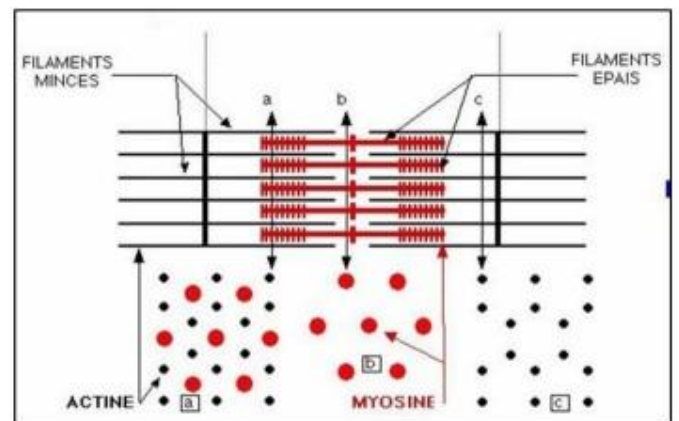
Striated muscle



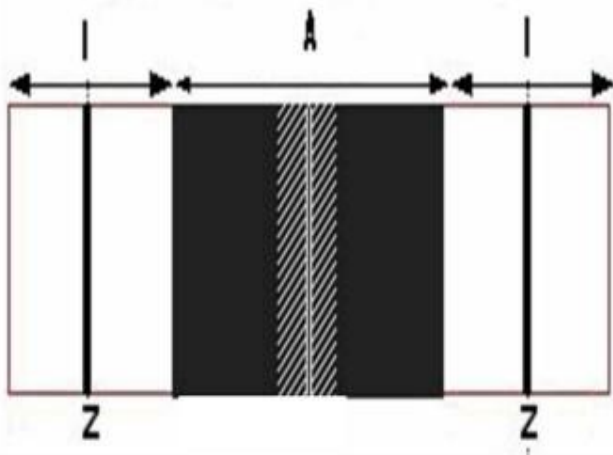
Skeletal muscle fibers



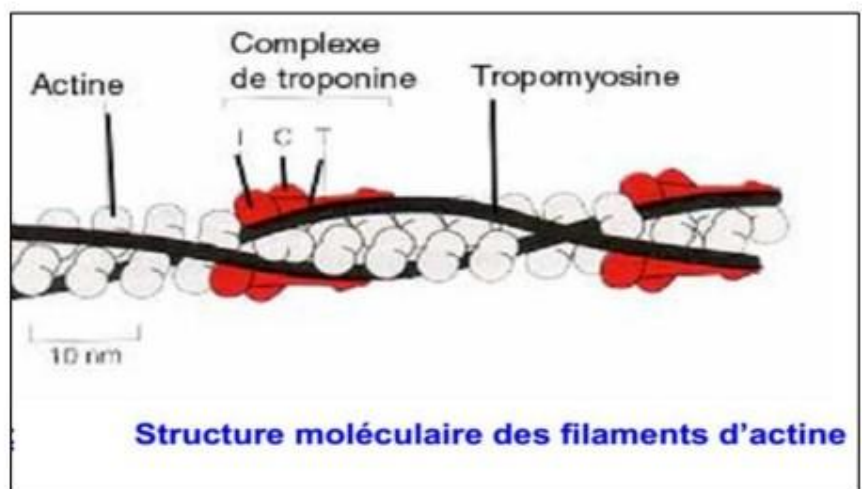
Structure moléculaire du filament de myosine



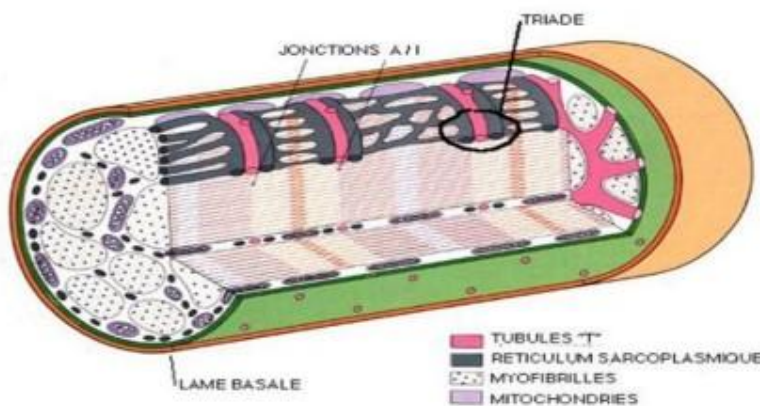
Sarcomère



Sarcomère



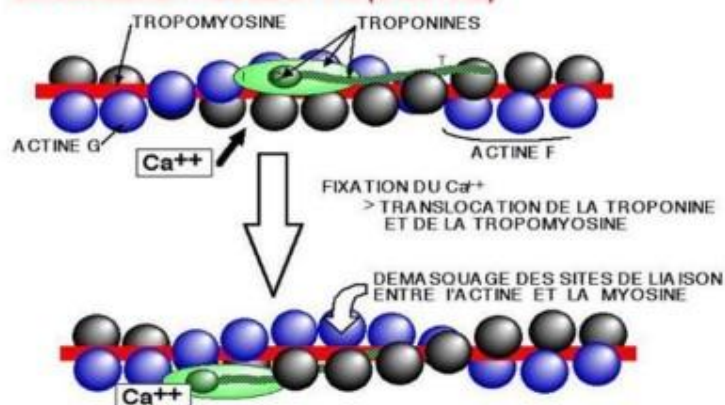
Structure moléculaire des filaments d'actine



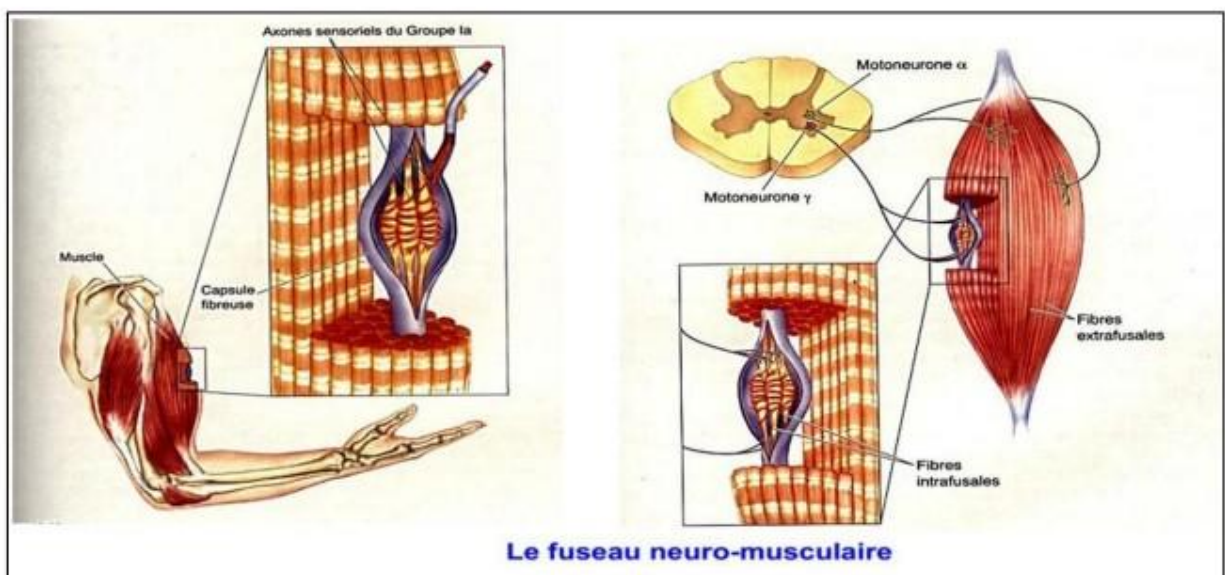
La triade est localisée approximativement dans le plan de jonction entre le disque A et le disque I

Organisation du réticulum endoplasmique lisse et des tubules T dans la cellule musculaire striée

SARCOMERE : FILAMENT FIN (D'ACTINE)



Microfilament d'actine et protéines associées



Le fuseau neuro-musculaire

